

Graph Puzzles III.1:

A proof of Sabidussi's compatibility conjecture with 4 colours

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ABSTRACT: We prove Sabidussi's compatibility conjecture. Let G be a finite connected multigraph in which every vertex has even degree and the minimum degree is at least four, and let T be an Euler tour of G . The edges of G can be partitioned into circuits (connected 2-regular subgraphs) so that no circuit contains two edges used consecutively anywhere in T . In fact, the edges can be four-coloured so that every such pair receives different colours and every colour class has even degree at every vertex. We use a counting argument based on the Chevalley–Warning theorem to show that a four-colouring with the required properties exists. Splitting each colour class into circuits then gives the desired compatible decomposition.

1 Introduction

All graphs in this paper are finite multigraphs; parallel edges are allowed. An edge has two distinct *half-edges*, each incident with a vertex. A graph is *even* if every vertex has even degree, and *Eulerian* if it is connected and even. A *circuit* is a nonempty connected 2-regular subgraph. A *circuit decomposition* is a partition of the edge set into circuits.

A transition at a vertex is an unordered pair of distinct half-edges incident with that vertex. A transition system pairs the half-edges at each vertex. A circuit pairs its two half-edges at each vertex that it meets; it is *compatible* with a transition system if none of these pairs is a prescribed transition. An Euler tour induces a transition system by pairing the arriving and departing half-edges at each passage through a vertex.

We prove the following strengthening of Sabidussi's conjecture.

Theorem 1.1. *Let G be a finite Eulerian multigraph with minimum degree $\delta(G) \geq 4$, and let*

$$T = (e_0, e_1, \dots, e_{m-1}), \quad m = |E(G)|,$$

be an Euler tour, with indices read modulo m . There is a colouring $\chi : E(G) \rightarrow \mathbb{F}_2^2$ such that

1. $\chi(e_{i-1}) \neq \chi(e_i)$ for every i ;
2. *every vertex has even degree in each colour class.*

Consequently, G has a circuit decomposition compatible with the transition system induced by T .

The conjecture, due to Gert Sabidussi, was recorded by Fleischner in [Fle80], where he also proved it for planar graphs. Fan and Zhang proved the conjecture for K_5 -minor-free graphs [FZ00]. Fleischner also showed that it suffices to consider graphs in which every vertex has degree four or six [Fle84, Lemma 1, p. 236]. The construction below treats every even degree directly, however it was inspired by the aforementioned reduction.

The proof constructs the edge colouring asserted in theorem 1.1. We begin with its two requirements: consecutive tour edges must have different colours, and each colour class must have even degree at every vertex. We express the second requirement as equations over $\mathbb{F}_2$, then use these equations together with the first requirement to develop the construction. A count modulo two using the Chevalley–Warning theorem will show that the required choices can always be made.

The theorem has also been formalized in Lean. The formalization is available in the author’s GitHub repository: <https://github.com/gexahedron/sabidussi-lean>.

1.1 Statement of AI use

The proof in this paper is essentially due to gpt 5.6 pro, produced by 2 iterations of a prompt similar to [Ope26a], with a help of a couple of additional hints.

1.2 Acknowledgements

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2 From compatible circuits to colour parity

We fix the finite Eulerian multigraph G , with $m = |E(G)|$ edges, and with an Euler tour $T = (e_0, e_1, \dots, e_{m-1})$. We begin by explaining the two requirements on the colouring.

Lemma 2.1. *Suppose that an edge colouring has even degree in each colour class at every vertex, and assigns different colours to consecutive edges of T . Then the colour classes can be decomposed into circuits that are compatible with the transition system induced by T .*

Proof. Every finite even multigraph decomposes into circuits: take an Euler tour in each component containing an edge and split it at repeated vertices until the resulting closed trails are circuits. Apply this separately to each colour class. The resulting circuits partition $E(G)$. Each is monochromatic, whereas the two half-edges of every prescribed transition belong to differently coloured edges. Thus none of its transitions is prescribed. $\square$

For a graph in which every vertex has degree four, alternating two colours along T already works. The tour has even length, and each of the two passages at a vertex

contributes one half-edge of each colour. At a degree-six vertex, the same alternation would contribute three half-edges of each colour, so it would fail the even-degree requirement. We use four colours, represented by the vectors in $\mathbb{F}_2^2$, to obtain more local choices.

All vector calculations from now on take place over $\mathbb{F}_2$. If every colour occurs evenly among the half-edges at a vertex, their vector sum is zero. The converse fails: the four distinct vectors $(0,0), (1,0), (0,1), (1,1)$ sum to zero, although each occurs once. We need one further bit to distinguish this possibility from even multiplicities. The function

$$q : \mathbb{F}_2^2 \longrightarrow \mathbb{F}_2, \quad q(a, b) = ab,$$

provides it: q is one exactly at the colour $(1,1)$.

Lemma 2.2 (Four-colour parity). *Let $(z_h)_{h \in H}$ be a family of vectors in $\mathbb{F}_2^2$, where $|H|$ is even. Every colour occurs an even number of times if and only if*

$$\sum_{h \in H} z_h = 0, \quad \sum_{h \in H} q(z_h) = 0. \quad (1)$$

Proof. Even multiplicities imply both equations. Conversely, let $n_{ab} \in \mathbb{F}_2$ be the parity of the multiplicity of (a, b) . The even cardinality and the two coordinates of the vector sum give

$$n_{00} + n_{10} + n_{01} + n_{11} = 0, \quad n_{10} + n_{11} = 0, \quad n_{01} + n_{11} = 0.$$

The quadratic sum is n_{11} . If it is zero, these equations successively give $n_{10} = n_{01} = n_{00} = 0$. $\square$

Thus our colouring problem has three precise requirements: consecutive colours must differ, the vector sum at each vertex must vanish, and the quadratic sum at each vertex must vanish.

3 Difference labels and the remaining obstruction

3.1 Local conditions and provisional integration

Let v_i be the vertex at which T passes from e_{i-1} to e_i , and put

$$I_v = \{i \in \{0, \dots, m-1\} : v_i = v\}.$$

The passages indexed by I_v pair all half-edges at v , hence $|I_v| = \frac{\deg_G(v)}{2} \geq 2$. We choose a starting point on T , so that comparisons $j < i$ use the ordinary linear order $0 < 1 < \dots < m-1$.

Write $x_i \in \mathbb{F}_2^2$ for the intended colour of e_i . At passage i , we require $x_{i-1} \neq x_i$. Over $\mathbb{F}_2^2$, this is equivalent to $x_{i-1} + x_i \neq 0$, so we introduce the *difference label* (we use "+" sign instead of "-" because over $\mathbb{F}_2^2$ they are equivalent):

$$y_i = x_{i-1} + x_i.$$

Each y_i records the change in colour between the two edges used at passage i . The distinctness requirement is now simply $y_i \neq 0$. The same variables also express the vector-sum requirement: the sum of the incident half-edge colours at v is

$$\sum_{i \in I_v} (x_{i-1} + x_i) = \sum_{i \in I_v} y_i.$$

The first two colouring requirements therefore suggest choosing labels with

$$y_i \neq 0 \quad (0 \leq i < m), \quad \sum_{i \in I_v} y_i = 0 \quad (v \in V(G)). \quad (2)$$

This choice is local: each vertex assigns one label to each of its passages, subject to a zero-sum condition. We will choose these differences first and recover the edge colours by following the tour. For the recovery to be consistent, the colour must return to its starting value after one full traversal. The local zero sums guarantee precisely this.

Lemma 3.1 (Integrating the differences). *Suppose the labels satisfy eq. (2). Set $x_{-1} = 0$ and*

$$x_i = \sum_{j=0}^i y_j \quad (0 \leq i < m). \quad (3)$$

Then $x_{m-1} = x_{-1}$, consecutive tour edges have different colours, and the sum of the incident half-edge colours is zero at every vertex.

Proof. The sets I_v partition the passage indices, so the local sums give $\sum_{i=0}^{m-1} y_i = 0$. Thus $x_{m-1} = 0 = x_{-1}$, and the colours close around the tour. The prefix formula gives $x_{i-1} + x_i = y_i \neq 0$, including at the passage with index zero. Finally, the incident half-edge colours at v are $(x_{i-1}, x_i)_{i \in I_v}$; their sum is the local sum of the y_i 's, which vanishes. $\square$

The chosen differences therefore produce an edge colouring with distinct consecutive colours and zero incident vector sums. It remains to choose them so that each colour occurs evenly at every vertex.

3.2 The quadratic obstruction

The remaining condition is that the number of incident half-edges coloured $(1, 1)$ should be even at every vertex. Since q is one exactly at this colour, its parity at v is $\sum_{i \in I_v} (q(x_{i-1}) + q(x_i))$. The lemma 2.2 says that every colour class has even degree at v if and only if this sum is zero. Our goal is to rewrite it in terms of the difference labels y_i instead of the colours x_i .

For $a = (a_1, a_2)$ and $b = (b_1, b_2)$ in $\mathbb{F}_2^2$, introduce the *polar form*

$$\langle a, b \rangle = a_1 b_2 + a_2 b_1.$$

This form is bilinear, symmetric, and alternating. Expanding $q(a + b)$ gives the polarization identity

$$q(a + b) = q(a) + q(b) + \langle a, b \rangle. \quad (4)$$

Using $x_i = x_{i-1} + y_i$ and the prefix formula eq. (3), we obtain

$$q(x_{i-1}) + q(x_i) = q(y_i) + \langle x_{i-1}, y_i \rangle = q(y_i) + \sum_{0 \leq j < i} \langle y_i, y_j \rangle.$$

Summing over $i \in I_v$, we separate the earlier passages at v from those at other vertices:

$$\begin{aligned} \sum_{i \in I_v} (q(x_{i-1}) + q(x_i)) &= \sum_{i \in I_v} q(y_i) + \sum_{\substack{i, j \in I_v \\ j < i}} \langle y_i, y_j \rangle \\ &\quad + \sum_{u \neq v} \sum_{\substack{i \in I_v, j \in I_u \\ j < i}} \langle y_i, y_j \rangle. \end{aligned}$$

The first two sums involve only labels at v . Repeated use of eq. (4) combines them into the quadratic value of their sum, which vanishes by the local zero-sum condition:

$$\sum_{i \in I_v} q(y_i) + \sum_{\substack{i, j \in I_v \\ j < i}} \langle y_i, y_j \rangle = q\left(\sum_{i \in I_v} y_i\right) = q(0) = 0. \quad (5)$$

Only terms pairing a passage at v with a passage at another vertex remain. For distinct vertices v, u , call their contribution the *interaction*

$$\beta_{vu}(y) = \sum_{\substack{i \in I_v, j \in I_u \\ j < i}} \langle y_i, y_j \rangle. \quad (6)$$

For each vertex v , define its total interaction with the other vertices by

$$Q_v(y) = \sum_{u \neq v} \beta_{vu}(y). \quad (7)$$

With this notation, the regrouping above gives

$$\sum_{i \in I_v} (q(x_{i-1}) + q(x_i)) = Q_v(y). \quad (8)$$

We have therefore expressed the parity of the incident $(1,1)$ -coloured half-edges entirely in terms of the difference labels. The remaining requirements are the *balancing equations*

$$Q_v(y) = 0 \quad (v \in V(G)). \quad (9)$$

4 Local assignments at each vertex

The preceding section reduced the colouring problem to choosing difference labels y_i satisfying three requirements:

$$\begin{aligned} y_i &\neq 0 & (0 \leq i < m), \\ \sum_{i \in I_v} y_i &= 0 & (v \in V(G)), \\ Q_v(y) &= 0 & (v \in V(G)). \end{aligned}$$

Nonvanishing makes consecutive tour edges have different colours. The local zero sums ensure that integration closes around the tour and that the incident vector sums vanish. The conditions $Q_v(y) = 0$ then give even multiplicities of all four colours.

We will give each vertex three assignments of its passage labels, each satisfying the first two requirements. The remaining task is to choose one assignment at each vertex so that the third requirement holds everywhere.

The labels in each assignment will be linear functions of a state $t_v \in \mathbb{F}_2^2$, with its three nonzero values giving the admissible choices. This makes the interactions bilinear and each Q_v a quadratic polynomial in the state coordinates. The count in section 5 will use Chevalley–Warning to show that the states can be chosen to meet the third requirement.

4.1 The local patterns

For the local construction, put

$$\Omega = \mathbb{F}_2^2 \setminus \{0\}, \quad \rho(r, s) = (s, r + s).$$

The map ρ cycles the three elements of Ω , and

$$t + \rho t + \rho^2 t = 0 \quad (t \in \mathbb{F}_2^2). \quad (10)$$

For two passages, the labels must be nonzero and sum to zero, so they have the form (t, t) , with $t \in \Omega$. For three passages, they must be the three distinct nonzero vectors. We use their three cyclic arrangements

$$(t, \rho t, \rho^2 t) \quad (t \in \Omega).$$

Thus both degrees four and six admit a chosen family of three patterns. At higher even degrees, we add pairs of equal labels to these patterns. This implements the same construction directly at each vertex.

Write $I_v = \{i_1 < \dots < i_k\}$, where $k \geq 2$. For every $t \in \mathbb{F}_2^2$, define $\Delta_{v,t} : I_v \rightarrow \mathbb{F}_2^2$ as follows. If k is even, put $\Delta_{v,t}(i_j) = t$ for all j . If k is odd, put

$$\Delta_{v,t}(i_j) = \begin{cases} t, & 1 \leq j \leq k-2, \\ \rho t, & j = k-1, \\ \rho^2 t, & j = k. \end{cases}$$

The admissible states are $t \in \Omega$. We extend the definition to $t = 0$ so that the patterns are linear maps defined on the whole vector space $\mathbb{F}_2^2$. The polynomial count in section 5 will first allow zero states and then exclude them.

A *pattern* is an assignment to the individual passages at a vertex. The state t_v selects the entire assignment Δ_{v,t_v} , and the resulting labels will be $y_i = \Delta_{v,t_v}(i)$ for $i \in I_v$.

Quite straightforwardly, we have the following lemma:

Lemma 4.1. *The patterns satisfy*

$$\Delta_{v,t}(i) \neq 0 \quad (i \in I_v, t \in \Omega), \quad (11)$$

$$\sum_{i \in I_v} \Delta_{v,t}(i) = 0 \quad (t \in \mathbb{F}_2^2). \quad (12)$$

For each fixed passage i , the label $\Delta_{v,t}(i)$ is linear in the state t .

4.2 Interactions between the states

For distinct v, u , substitute the patterns into the interaction $\beta_{vu}(y)$ from eq. (6). We write it in terms of the two states as

$$\beta_{vu}(t, s) = \sum_{\substack{i \in I_v, j \in I_u \\ j < i}} \langle \Delta_{v,t}(i), \Delta_{u,s}(j) \rangle \quad (t, s \in \mathbb{F}_2^2). \quad (13)$$

Apply same substitution $y_i = \Delta_{v,t_v}(i)$, $i \in I_v$, in $Q_v(y)$ from eq. (7). We denote the resulting expression in the state variables by $Q_v(t)$:

$$Q_v(t) = \sum_{u \neq v} \beta_{vu}(t_v, t_u). \quad (14)$$

The requirements $Q_v(y) = 0$ thus become $Q_v(t) = 0$: the remaining colouring obstruction is now a condition on the state choices.

Lemma 4.2. *The maps β_{vu} are bilinear and satisfy*

$$\beta_{vu}(t, s) = \beta_{uv}(s, t) \quad (t, s \in \mathbb{F}_2^2). \quad (15)$$

Proof. Linearity of the patterns in their states and bilinearity of the pairing give bilinearity.

For transpose symmetry, each passage belongs to exactly one vertex, so I_v and I_u are disjoint when $v \neq u$. Every pair of passages $i \in I_v, j \in I_u$ therefore occurs in exactly one of the orders $j < i$ or $i < j$. Using the symmetry of the pairing, we obtain

$$\begin{aligned} \beta_{vu}(t, s) + \beta_{uv}(s, t) &= \sum_{i \in I_v} \sum_{j \in I_u} \langle \Delta_{v,t}(i), \Delta_{u,s}(j) \rangle \\ &= \left\langle \sum_{i \in I_v} \Delta_{v,t}(i), \sum_{j \in I_u} \Delta_{u,s}(j) \right\rangle = 0. \end{aligned}$$

The last equality uses eq. (12) at both vertices. $\square$

Each vertex now has three admissible nonzero states, and the interactions extend to bilinear maps with transpose symmetry on $\mathbb{F}_2^2$. We use this extension to count choices satisfying all the balancing equations simultaneously.

5 Applying Chevalley–Warning

The Chevalley–Warning theorem [Che35, War35] says that polynomials over $\mathbb{F}_2$ in n scalar variables have an even number of common zeros when the sum of their degrees is less than n .

To use this theorem, we must allow each state to range over all of $\mathbb{F}_2^2$. The added zero states give invalid local patterns, so a count on the whole vector space alone does not settle our problem. We will count balanced families with specified states set to zero, then use inclusion–exclusion to retain only families in which every state is nonzero.

Lemma 5.1. *Let U be a finite set. For each pair of distinct elements u, v , let $\beta_{uv} : \mathbb{F}_2^2 \times \mathbb{F}_2^2 \rightarrow \mathbb{F}_2$ be bilinear and satisfy*

$$\beta_{uv}(r, s) = \beta_{vu}(s, r).$$

For a family $t = (t_u)_{u \in U}$, put

$$Q_u(t) = \sum_{v \neq u} \beta_{uv}(t_u, t_v).$$

The number of families with $t_u \in \Omega = \mathbb{F}_2^2 \setminus \{0\}$ and $Q_u(t) = 0$ for every $u \in U$ is odd. In particular, such a family exists.

Proof. Allow the states to range over $\mathbb{F}_2^2$. By bilinearity, each Q_u is a polynomial of degree at most two in their scalar coordinates. Transpose symmetry gives the polynomial identity

$$\sum_{u \in U} Q_u(t) = 0, \tag{16}$$

because every unordered pair contributes twice. This makes one balancing equation redundant.

For $S \subseteq U$, let $F(S)$ be the integer number of balanced families with $t_u = 0$ for $u \notin S$; states in S may be any of the four vectors. The equations outside S vanish identically by bilinearity. If $S \neq \emptyset$, the remaining equations still sum to zero by eq. (16), so one can be omitted. There are at most $|S| - 1$ quadratic equations in $2|S|$ scalar variables, with degree bound

$$2(|S| - 1) < 2|S|.$$

The inequality verifies the degree condition in Chevalley–Warning, which implies that $F(S)$ is even for every nonempty S . When $|S| = 1$, there are no equations left and all four states are solutions. For $S = \emptyset$, only the all-zero family is possible, and it is balanced, so $F(\emptyset) = 1$.

Let N be the number of balanced families with every state nonzero. Inclusion–exclusion, applied to the conditions $t_u = 0$, gives

$$\begin{aligned} N &= \sum_{S \subseteq U} (-1)^{|U| - |S|} F(S) \\ &\equiv \sum_{S \subseteq U} F(S) \equiv F(\emptyset) \equiv 1 \pmod{2}. \end{aligned} \quad \square$$

6 Completing the colouring

All the ingredients now have the roles for which they were constructed. The admissible assignments give nonzero labels that sum to zero at each vertex; balancing removes the quadratic obstruction; integration turns the resulting labels into edge colours.

Proof of theorem 1.1. The interactions in eq. (13) are bilinear and symmetric by lemma 4.2. Applying lemma 5.1 with $U = V(G)$ gives a family $t = (t_v)_{v \in V(G)}$, with $t_v \in \Omega$, such that $Q_v(t) = 0$ for every v . Put

$$y_i = \Delta_{v,t_v}(i) \quad (i \in I_v).$$

By lemma 4.1, these labels satisfy eq. (2). Define x_i by eq. (3) and set $\chi(e_i) = x_i$. Lemma 3.1 shows that the colours close around T , consecutive tour edges have different colours, and the incident vector sum vanishes at every vertex.

By eq. (14), $Q_v(y) = Q_v(t) = 0$. Applying eq. (8) gives zero incident quadratic sum at every vertex. Lemma 2.2 now gives even degree in every colour class. Finally, lemma 2.1 decomposes the colour classes into compatible circuits. $\square$

7 A consequence for cycle double covers

For this consequence, a *cycle* means an even subgraph and need not be connected. A *k-cycle double cover* is a family of at most k cycles in which every edge occurs exactly twice. A circuit is *dominating* if every edge of the graph has at least one endpoint on it. The cycle double cover construction of OpenAI [Ope26b] gives eight even subgraphs indexed by $\mathbb{F}_2^3$. In the presence of a prescribed dominating circuit, theorem 1.1 improves the bound and retains that circuit as a member.

Corollary 7.1. *Let H be a finite cubic graph and let C be a dominating circuit of H . Then H has a 5-cycle double cover with C as a member.*

Proof. Every vertex of C has exactly one incident edge outside C . Since C is dominating, every vertex outside C has all three neighbours on C . Hence each component of $H - E(C)$ is either a single edge with both ends on C , or a star with its three leaves on C . Contract these components, retaining the edges of C , to obtain a multigraph G [Fle84, pp. 236–237]. The contracted vertices have degree four or six, and traversing C once gives an Euler tour T of G . Apply theorem 1.1 to obtain a four-colouring χ of $E(G) = E(C)$.

Fix a colour $z \in \mathbb{F}_2^2$. At every vertex of C , the two edges of C are consecutive in T and have different colours. For a component K of $H - E(C)$, the number of its vertices incident with a z -coloured edge of C is therefore the degree of the contracted vertex in colour z . This number is even and at most three, so it is zero or two.

Let D_z consist of the z -coloured edges of C , together with the unique path in each component K joining its two such vertices, whenever they exist. At a vertex of C , the degree in D_z is two if that vertex has an incident z -coloured edge of C , and

zero otherwise. A vertex outside C has degree two if one of the added paths passes through it, and zero otherwise. Thus D_z is an even subgraph of H .

Every edge of C belongs to exactly one D_z . For an edge f outside C , choose an end u on C . The edge f belongs to D_z exactly when u is incident with a z -coloured edge of C , since u is a leaf of its component. The two edges of C at u have different colours, so f belongs to exactly two of the D_z . Adding C gives the second covering of its own edges, hence the required 5-cycle double cover. $\square$

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